

1. Introduction to Machine Learning

Machine Learning is a subfield of Artificial Intelligence that allows computers to **learn from data and improve their performance automatically without being explicitly programmed**.

In traditional programming, a programmer writes **rules and instructions** for the computer. But in machine learning, the computer **learns patterns from data** and uses those patterns to make predictions or decisions.

Machine learning systems work by analyzing large amounts of data, identifying patterns, and building models that can make predictions on new or unseen data.

Basic Working of Machine Learning

The general process of machine learning includes:

1. **Data Collection** – Collect relevant data from different sources.
2. **Data Preprocessing** – Clean and prepare the data for analysis.
3. **Model Training** – Use algorithms to train the model using the dataset.
4. **Model Testing** – Evaluate the model using test data.
5. **Prediction or Decision Making** – Use the trained model to make predictions.

Example

Suppose we want to predict **house prices**.

We give the system data such as:

- Size of the house
- Number of rooms
- Location
- Age of the house

The machine learning model analyzes these patterns and **predicts the price of a new house**.

2. Types of Learning in Machine Learning

Machine learning algorithms are mainly divided into three types depending on **how the data is provided and how the system learns**.

1. Supervised Learning

In **Supervised Learning**, the model is trained using **labeled data**.

This means that both the **input data and the correct output** are provided to the model during training.

The algorithm learns the relationship between input and output and then predicts the output for new data.

Example

If we want to classify emails as **Spam or Not Spam**, we train the system with emails that are already labeled.

Example dataset:

Email Text	Output
Buy now offer	Spam
Meeting schedule	Not Spam

The model learns the pattern and predicts whether new emails are spam.

Common Algorithms

Some popular supervised learning algorithms include:

- Linear Regression
- Logistic Regression
- Decision Tree
- K-Nearest Neighbour (KNN)
- Support Vector Machine (SVM)
- Random Forest

Applications

- House price prediction
- Email spam detection
- Medical diagnosis

2. Unsupervised Learning

In **Unsupervised Learning**, the model is trained using **unlabeled data**.

This means the system receives only input data without any predefined output.

The algorithm tries to **discover hidden patterns or relationships** in the data.

Example

Suppose a company has customer purchase data but no categories.

The algorithm groups customers based on **similar buying behavior**.

These groups are called **clusters**.

Common Algorithms

Some common unsupervised learning algorithms are:

- K-Means Clustering
- Hierarchical Clustering
- Gaussian Mixture Model
- Association Rule Mining
- Apriori Algorithm

Applications

- Customer segmentation
 - Market basket analysis
 - Data compression
-

3. Reinforcement Learning

In **Reinforcement Learning**, an agent learns by interacting with the environment through **trial and error**.

The system performs actions and receives **rewards or penalties** based on the results.

The main goal is to **maximize the total reward**.

Main Components

1. **Agent** – the learner or decision maker
2. **Environment** – the system in which the agent operates
3. **Action** – the choices made by the agent
4. **Reward** – feedback from the environment

Example

A robot learning to walk:

- If it walks correctly → reward
- If it falls → penalty

Over time, the robot learns the best way to walk.

Applications

- Game playing AI
 - Robotics
 - Self-driving cars
 - Resource management systems
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3. Applications of Machine Learning

Machine learning is widely used in many industries and real-world systems.

1. Recommendation Systems

Machine learning is used to recommend products, movies, or content based on user behavior.

For example:

- Netflix recommends movies based on your watch history.
 - Amazon recommends products based on previous purchases.
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2. Image Recognition

Machine learning can identify objects, faces, or patterns in images.

Applications include:

- Face recognition systems
 - Medical image analysis
 - Security surveillance systems
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3. Speech Recognition

Speech recognition systems convert **spoken language into text**.

Used in voice assistants such as:

- Google Assistant
- Amazon Alexa

These systems understand voice commands and perform tasks.

4. Email Spam Detection

Email services like Gmail use machine learning algorithms to identify spam emails by analyzing patterns in email content.

5. Fraud Detection

Banks and financial institutions use machine learning to detect **fraudulent transactions** in credit cards and online payments.

The system analyzes transaction patterns and flags suspicious activities.

6. Self-Driving Cars

Companies such as Tesla use machine learning to detect roads, pedestrians, traffic signals, and obstacles to drive vehicles automatically.

Linear Regression Model

Linear Regression is a supervised machine learning algorithm used to **predict a continuous value** based on the relationship between variables.

It finds a **linear relationship (straight line)** between:

- **Independent variable (input)** $\rightarrow x$
- **Dependent variable (output)** $\rightarrow y$

The goal of linear regression is to **predict the value of y when a new value of x is given**.

Linear Regression Equation

$$y = mx + b$$

m

b

-10-8-6-4-2246810-10-5510y-interceptx-intercept

Where:

- **y** = Predicted output (dependent variable)
- **x** = Input variable (independent variable)
- **m** = Slope of the line (relationship strength)
- **b** = Intercept (value of y when $x = 0$)

This equation represents the **best-fit straight line** that minimizes prediction error.

Example

Suppose we want to predict **house price** based on **house size**.

House Size (sq ft) Price

1000	20 lakh
1500	30 lakh
2000	40 lakh

Linear regression will find a line that best fits these points and then **predict the price of a new house**.

Example:

If house size = **1800 sq ft**, the model predicts the price based on the learned line.

Types of Linear Regression

1. Simple Linear Regression

Uses **one independent variable**.

Example:

- Predict salary based on **years of experience**.

Equation:

$$y = mx + b$$

2. Multiple Linear Regression

Uses **multiple input variables**.

Example:

Predict house price using:

- house size
- number of rooms
- location

Equation:

$$y = b + m_1x_1 + m_2x_2 + m_3x_3$$

Steps in Linear Regression

1. Collect the dataset
2. Plot the data points
3. Find the **best-fit line**
4. Train the model

5. Predict new values

The algorithm chooses a line that **minimizes prediction error**.

Applications of Linear Regression

Linear regression is widely used in real life.

Examples:

- **House price prediction**
 - **Sales forecasting**
 - **Stock market trend analysis**
 - **Risk assessment in finance**
 - **Demand prediction**
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Advantages

- Simple and easy to understand
 - Fast to train
 - Works well with linear relationships
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Limitations

- Works only when the relationship is **linear**
 - Sensitive to **outliers**
 - Cannot capture complex patterns
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Naïve Bayes Classifier

The **Naïve Bayes Classifier** is a **supervised machine learning algorithm** used mainly for **classification problems**.

It is based on the mathematical rule called **Bayes' Theorem**, which calculates the probability of an event based on prior knowledge of related conditions.

The word “**Naïve**” means that the algorithm assumes all features (inputs) are **independent of each other**, which is a very simple assumption.

Even though this assumption is not always true, the algorithm works **very well in many real-world problems**.

Bayes Theorem Formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)} \approx 0.81, P(B) \approx 0.35$$

$$P(B)=0.35P(B|A)P(A)=0.28P(A|B)\sim0.81\text{Posterior} = \text{useful evidence} / \text{total evidence}$$

Where:

- **P(A|B)** = Posterior probability (probability of A given B)
- **P(B|A)** = Likelihood
- **P(A)** = Prior probability of event A
- **P(B)** = Evidence or total probability of B

The classifier uses this formula to **calculate the probability of each class** and chooses the class with the **highest probability**.

Example (Spam Email Detection)

Suppose we want to classify emails as:

- **Spam**

- **Not Spam**

If an email contains words like:

- "Free"
- "Offer"
- "Discount"

The algorithm calculates the probability that the email belongs to the **spam category**.

If the spam probability is higher, the email will be classified as **Spam**.

Email services such as Gmail use this technique for spam filtering.

Types of Naïve Bayes

1. Gaussian Naïve Bayes

Used when the features are **continuous values**.

Example:

- height
 - weight
 - temperature
-

2. Multinomial Naïve Bayes

Used mainly for **text classification**.

Example:

- spam detection
 - sentiment analysis
 - document classification
-

3. Bernoulli Naïve Bayes

Used when features are **binary (0 or 1)**.

Example:

- word present or not present in a document.

Steps of Naïve Bayes Algorithm

1. Calculate the **prior probability** for each class.
 2. Calculate the **likelihood probability** of each feature.
 3. Apply **Bayes theorem** to compute posterior probability.
 4. Choose the class with the **highest probability**.
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Applications of Naïve Bayes

Naïve Bayes is widely used in many applications:

1. **Email Spam Detection**
 2. **Sentiment Analysis**
 3. **Document Classification**
 4. **Recommendation Systems**
 5. **Medical Diagnosis**
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Advantages

- Simple and easy to implement
 - Works well with **large datasets**
 - Very **fast training and prediction**
 - Works well for **text classification**
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Disadvantages

- Assumes **independent features**, which may not always be true
- Performance may reduce if features are highly correlated

Decision Tree

A **Decision Tree** is a **supervised machine learning algorithm** used for **classification and regression problems**.

It works like a **tree structure**, where the data is split into branches based on different conditions until a final decision is reached.

The model looks similar to a **flowchart**, where each internal node represents a decision and each branch represents the result of that decision.

Structure of a Decision Tree

A decision tree consists of three main parts:

1. Root Node

The **topmost node** of the tree.

It represents the **entire dataset** and starts the decision-making process.

2. Internal Node

These nodes represent **conditions or tests** on features.

Example:

- Is age > 30?
- Is income high?

3. Leaf Node

The **final node** that gives the **prediction or output**.

Example:

- Approve loan
 - Reject loan
-

Example of Decision Tree

Suppose a bank wants to decide whether to **approve a loan**.

Decision process:

1. Check **Income**
2. Check **Credit Score**

3. Check **Employment Status**

Example flow:

Income > 50,000 → Check credit score

Credit score good → **Loan Approved**

So the algorithm creates a **tree-like structure** of decisions.

How Decision Tree Works

The algorithm follows these steps:

1. Select the **best feature** to split the data.
2. Divide the dataset into **subsets**.
3. Repeat the process for each subset.
4. Stop when a **final decision (leaf node)** is reached.

To choose the best feature, the algorithm uses measures such as:

- **Information Gain**
 - **Entropy**
 - **Gini Index**
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Important Concepts

Entropy

Entropy measures the **impurity or randomness in the dataset**.

Entropy = $-\sum p_i \log_2(p_i)$

Lower entropy means **more pure data**.

Information Gain

Information Gain measures **how much information a feature provides** when splitting the data.

The feature with **highest information gain** becomes the decision node.

Applications of Decision Tree

Decision trees are widely used in real-world applications such as:

- **Loan approval systems**
 - **Medical diagnosis**
 - **Customer segmentation**
 - **Fraud detection**
 - **Risk analysis**
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Advantages

- Easy to understand and visualize
 - Works with both **numerical and categorical data**
 - Requires little data preprocessing
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Disadvantages

- Can **overfit** the training data
 - May become complex with large datasets
 - Small changes in data can change the tree structure
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K-Nearest Neighbour (KNN)

K-Nearest Neighbour (KNN) is a **supervised machine learning algorithm** used for **classification and regression problems**.

The main idea of KNN is **similar data points are located close to each other**.

When a new data point appears, the algorithm finds the **K closest data points** and predicts the result based on them.

Here **K** represents the **number of nearest neighbors** considered.

Working of KNN Algorithm

The KNN algorithm works in the following steps:

1. Choose the value of **K** (number of neighbors).
 2. Calculate the **distance** between the new data point and all other points in the dataset.
 3. Select the **K nearest data points**.
 4. Determine the **majority class** among those neighbors.
 5. Assign that class to the new data point.
-

Distance Calculation

To find the nearest neighbors, KNN uses a distance formula called **Euclidean Distance**.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

-10 -8 -6 -4 -2 2 4 6 8 10
-10 -5 0 5 10
(6.0, 6.0) (-6.0, -6.0)
 $d = 16.97$

Where:

- x_1, y_1 = coordinates of first point
- x_2, y_2 = coordinates of second point
- d = distance between the points

The smaller the distance, the **closer the neighbor**.

Example

Suppose we want to classify a **fruit** based on its features:

Weight Texture Fruit

150g Smooth Apple

170g Smooth Apple

130g Rough Orange

If a new fruit has weight **160g**, the algorithm finds the **nearest fruits**.

If most neighbors are **Apple**, the model predicts **Apple**.

Types of KNN Problems

1. Classification

Predicts **categories or labels**.

Example:

- Spam or Not Spam
 - Disease detection
-

2. Regression

Predicts **continuous values**.

Example:

- Predicting house price
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Choosing the Value of K

- **Small K** → sensitive to noise
- **Large K** → smoother prediction but may reduce accuracy

Usually **odd numbers (3, 5, 7)** are chosen to avoid ties.

Applications of KNN

KNN is used in many real-world applications:

1. **Recommendation systems**

2. **Image recognition**
 3. **Pattern recognition**
 4. **Medical diagnosis**
 5. **Credit scoring**
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Advantages

- Simple and easy to understand
 - No training phase required
 - Works well with small datasets
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Disadvantages

- Slow with large datasets
 - Requires high memory
 - Sensitive to irrelevant features
-

Short Exam Definition

K-Nearest Neighbour is a supervised machine learning algorithm that classifies new data points by finding the K closest data points in the dataset and assigning the majority class.

Logistic Regression

Logistic Regression is a **supervised machine learning algorithm** used for **classification problems**.

Even though the name contains “**regression**”, it is mainly used to **predict categorical outputs**, such as:

- Yes / No
- True / False
- Spam / Not Spam
- Disease / No Disease

Logistic regression predicts the **probability that a data point belongs to a particular class**.

The output value is always **between 0 and 1**, which represents probability.

Logistic Regression Function

Logistic regression uses a function called the **Sigmoid Function**.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

This function converts any value into a **probability between 0 and 1**.

Example

If the output probability is:

- **> 0.5 → Class 1 (Yes / True)**
 - **< 0.5 → Class 0 (No / False)**
-

Example of Logistic Regression

Suppose we want to predict **whether a student will pass or fail** based on the number of hours studied.

Study Hours Result

2	Fail
4	Fail

Study Hours Result

6 Pass

8 Pass

Logistic regression learns from this data and predicts the **probability of passing** for a new student.

Example:

Study = 5 hours

Prediction = **0.65 probability of passing**

Since it is greater than **0.5**, the result will be **Pass**.

Types of Logistic Regression

1. Binary Logistic Regression

Used when there are **two possible outcomes**.

Example:

- Spam / Not Spam
- Pass / Fail

2. Multinomial Logistic Regression

Used when there are **more than two classes**.

Example:

- Predicting type of fruit (Apple, Banana, Orange)

3. Ordinal Logistic Regression

Used when categories have **ordered values**.

Example:

- Ratings (Low, Medium, High)

Applications of Logistic Regression

Logistic regression is used in many practical fields:

- **Medical diagnosis** (disease prediction)
 - **Email spam detection**
 - **Customer churn prediction**
 - **Credit risk assessment**
 - **Marketing prediction**
-

Advantages

- Simple and easy to implement
 - Works well for **binary classification problems**
 - Requires less computational power
-

Disadvantages

- Works best only with **linear relationships**
 - Cannot handle very complex datasets well
 - Sensitive to outliers
-

Support Vector Machine (SVM)

Jarvis, here is a **clear explanation of Support Vector Machine (SVM)** that is good for **understanding and exams**.

Support Vector Machine (SVM)

A **Support Vector Machine (SVM)** is a **supervised machine learning algorithm** used for **classification and regression problems**.

The main goal of SVM is to **find the best boundary that separates different classes of data**.

This boundary is called a **hyperplane**.

SVM tries to choose the hyperplane that **maximizes the distance between the nearest data points of different classes**.

Important Concepts in SVM

1. Hyperplane

A **hyperplane** is a decision boundary that separates data into different classes.

Example:

If we have two classes like:

- Red points
- Blue points

The SVM algorithm draws a **line (in 2D)** or **plane (in higher dimensions)** that separates them.

2. Support Vectors

The data points that are **closest to the hyperplane** are called **support vectors**.

These points are very important because they **determine the position of the hyperplane**.

If these points move, the hyperplane also changes.

3. Margin

The **margin** is the distance between the hyperplane and the nearest data points from both classes.

SVM tries to **maximize this margin**.

A larger margin means better classification.

Example

Suppose we want to classify emails into:

- Spam
- Not Spam

The SVM algorithm finds the **best boundary** that separates spam emails from normal emails based on their features.

Types of SVM

1. Linear SVM

Used when the data can be separated using a **straight line**.

Example: simple classification problems.

2. Non-Linear SVM

Used when the data cannot be separated with a straight line.

In this case, SVM uses a technique called the **Kernel Trick** to transform data into a higher dimension where separation becomes easier.

Common kernels include:

- Linear Kernel
 - Polynomial Kernel
 - Radial Basis Function (RBF) Kernel
-

Applications of SVM

SVM is used in many real-world applications such as:

- **Image classification**

- **Text classification**
 - **Handwriting recognition**
 - **Face detection**
 - **Bioinformatics**
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Advantages

- Works well with **high-dimensional data**
 - Effective for **classification problems**
 - Good accuracy with small datasets
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Disadvantages

- Training can be **slow with very large datasets**
 - Choosing the correct **kernel** can be difficult
 - Requires careful parameter tuning
-

Random Forest Algorithm

The **Random Forest Algorithm** is a **supervised machine learning algorithm** used for **classification and regression problems**.

It works by creating **multiple decision trees** and combining their results to produce a **more accurate and stable prediction**.

Instead of relying on a single tree, Random Forest builds a **forest of many decision trees**, and the final prediction is made by **majority voting (classification)** or **average prediction (regression)**.

The name **Random Forest** comes from the idea of using **many randomly created decision trees**.

Working of Random Forest

The algorithm follows these steps:

1. Select **random samples from the dataset**.
2. Build a **decision tree for each sample**.
3. Each tree makes its **own prediction**.
4. The final result is determined by **majority voting (classification)** or **average value (regression)**.

Example:

Suppose 5 trees predict:

- Tree 1 → Apple
- Tree 2 → Apple
- Tree 3 → Orange
- Tree 4 → Apple
- Tree 5 → Orange

Final prediction → **Apple** (majority vote)

Why Random Forest is Better than a Single Decision Tree

A single decision tree may **overfit the data**, meaning it works well on training data but poorly on new data.

Random Forest solves this problem by:

- Using **multiple trees**
- Combining predictions
- Reducing **overfitting**

This makes the model **more accurate and reliable**.

Key Features of Random Forest

1. Uses **multiple decision trees**
 2. Uses **random subsets of data**
 3. Combines results using **voting or averaging**
 4. Improves **accuracy and stability**
-

Applications of Random Forest

Random Forest is widely used in real-world problems such as:

- **Medical diagnosis**
 - **Fraud detection in banking**
 - **Stock market prediction**
 - **Customer behavior prediction**
 - **Image classification**
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Advantages

- High accuracy
 - Reduces overfitting
 - Works well with large datasets
 - Handles missing data well
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Disadvantages

- Requires more **computational power**

- Slower than simple algorithms
 - Model can become complex
-

✔ Short Exam Definition

Random Forest is a supervised machine learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

✔ Jarvis, now your **Unit-1** is completely finished:

1. Introduction of Machine Learning
 2. Types of Learning
 3. Applications of Machine Learning
 4. Linear Regression
 5. Naïve Bayes
 6. Decision Tree
 7. K-Nearest Neighbour
 8. Logistic Regression
 9. Support Vector Machine
 10. Random Forest
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